Ruixin(Jack) Shi

Seattle, WA | xxruixin@gmail.com | 617-784-1401 | Portfolio | LinkedIn | GitHub

EDUCATION

Northeastern University, Seattle, WA — *May 2024 – Jan 2027 (Expected)*
Master of Science in Computer Science
Related courses: Machine Learning, Distributed Systems, Mobile App Development, Assistive Robotics, Advanced Software Development

Boston University, Boston, MA — *Sep 2020 – May 2024*
Bachelor of Arts in Computer Science
Related courses: Data Structures, Software Engineering, Networks, Artificial Intelligence, Computer Graphics, Operating Systems

EXPERIENCE

TikTok, Software Engineer Intern – Mobile Infrastructure, *San Jose, CA* — *Jun 2026 – Sep 2026*

- Shift TikTok's **internal debugging tool** for **Android and iOS** from GUI-based to **agent-driven**, letting developers debug with any AI agent via skills; shipped it and sped up development on both platforms.
- Owned the **Android agent proxy end to end**: **regex-based URL rerouting** to a local server, with a **debug-only hook** to pass security gates.
- Shipped **4 features** to the tool used by **500+ internal developers**: debugger auto-focus, proxy cache hardening, and page-metadata restore on reload; fixed a legacy device–desktop sync bug.
- Resolved **12+ cross-platform on-call incidents** and built an **AI Q&A bot** (ByteFaaS + team docs) with cited answers; its architecture was adopted for later team tooling.

Air Health.co, Software Engineer Lead (Part-time), *Remote* — *Dec 2025 – Jul 2026*

- Led a **3-engineer team** building a native mobile app in Swift (iOS) and Kotlin (Android) from requirements to production; **owned the full stack** — every client feature and the backend behind it.
- Migrated the backend from **Vercel + Cloudflare + Supabase to AWS** (Lambda, EC2, S3, SES), consolidating **three vendors into one** and reducing cold-start latency and monthly operating cost.
- Integrated a **real-time LLM** with an **on-device text-to-speech (TTS)** pipeline, delivering dynamic voice on iOS and Android with sub-second latency.

Northeastern University, Khoury College of Computer Sciences, Graduate Researcher (Dr. Yifan Hu), *Seattle, WA* — *May 2025 – Present*

- **Co-authored and presented (oral)** a paper at **Canadian AI 2026 / PMLR**; built a **1.4M-record pipeline** from the Open Academic Graph and fine-tuned **multilingual BERT (NameBERT)**, outperforming SOTA baselines (NamePrism, EthnicSeer, Name2Nat).
- Designed an **LLM-based data augmentation pipeline** for long-tail countries (**~17%** tail Macro-F1 gain); benchmarked vs. GPT-5 / Claude / Gemini at **~400x throughput** (1,831 names/sec) and near-zero per-query cost.

Framo Media L.L.C., AI Development Intern, *Seattle, WA* — *Sep 2025 – Dec 2025*

- Integrated **AI agent tools** (ComfyUI, OpenRouter API) into production media workflows, automating multi-step content generation; deployed and administered a **PostgreSQL** server for multi-user asset collaboration.

PROJECT EXPERIENCE

cowork — Multi-Agent Coordination Kit for Parallel Sessions, *Bash, Claude Code, Git* — *Sep 2026*

- Designed a **file-based lead/seat protocol** that lets one lead session plan, review, and integrate while N seat sessions implement in parallel — atomic **mkdir locks**, **mtime-watcher wake-ups**, and automatic **takeover of dead sessions**, no network or server required.
- Shipped a CLI (**init / grow / tree / update**) that recurses to **nested teams of any depth**, plus a Claude Code skill and an end-to-end smoke test.

Capstone Showcase Website, *React, TypeScript, Sanity.io, Vercel* — *Sep 2025 – Dec 2025*

- Co-built and deployed a **full-stack web platform** for Northeastern capstone projects — **React + TypeScript** frontend, **Sanity.io** headless CMS, **Vercel CI/CD**, multi-user content management.

Fit-Mirror — On-Device AI Fitness Coach, *Computer Vision, On-Device AI* — *Mar 2025*

- Led a team to build a **real-time computer-vision system** that tracks body posture and projects live corrective feedback, running fully **on-device**; won **Popular Choice and Judges' Choice** at the Qualcomm & Microsoft On-Device AI Hackathon.

UnWire — Mobile App for Local Network Chat, *Flutter, Dart* — *Sep 2024 – Dec 2024*

- Built a **Flutter mobile app** enabling real-time peer-to-peer chat over local networks with no internet connectivity; designed the full application architecture and core messaging features end-to-end.

PUBLICATIONS & AWARDS

- C. Ming, **R. Shi**, Y. Hu. "NameBERT: Scaling Name-Based Nationality Classification with LLM-Augmented Open Academic Data." **Canadian AI 2026 / PMLR** (Oral Presentation).
- **Qualcomm & Microsoft On-Device AI Hackathon — Winner** (Team Leader, Northeastern University, Mar 2025) — Popular Choice and Judges' Choice awards for Fit-Mirror.

TECHNICAL SKILLS

- **Languages:** Python, Java, Kotlin, Swift, JavaScript, TypeScript, Go, C++, SQL, Dart
- **Mobile & Frontend:** Android (Kotlin), iOS (Swift), Flutter, React, Next.js, Electron, HTML5/CSS3, Tailwind CSS
- **Backend & Cloud:** Spring Boot, Node.js, Express, Django, Flask, REST/GraphQL, WebSocket, Microservices, AWS (Lambda, EC2, S3, SES), ByteFaaS, Docker, Kubernetes, GitHub Actions, CI/CD, Linux
- **AI / ML & Agents:** LLM integration, RAG, MCP, agent tooling, fine-tuning, NLP, PyTorch, Transformers, scikit-learn, OpenRouter API, ComfyUI
- **Data & Testing:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch, Git, pytest, JUnit, Jest, Postman